

Table of Contents

Introduction.....	3
Question 1.....	3
Question 2.....	5
Question 3.....	7
Question 4.....	9
Question 5.....	10
Question 6.....	16
Question 7.....	17
Question 8.....	25
Conclusion.....	43

Introduction

The purpose of this project was to organize, calculate and illustrate data related to the ice concentrations of the Great Lakes. There are three data sets: one for 2023-2024, one for 2022-2023, and one for 2021-2022. This project includes importing raw data, performing intricate statistical calculations, and generating plots using the C programming language and GNUPlot. Utilizing these sources, this project aims to identify trends, compare and contrast seasonal changes, and compute important values such as annual averages and maximum ice coverage for each data set.

Analyzing and visualizing this data is extremely important for highlighting notable changes in ice concentration levels over the years. These drastic changes can directly relate to the increasing temperatures from climate change, emphasizing the negative impacts of human-generated greenhouse gas emissions on our environment. With this, the project will also demonstrate the effectiveness of C and GNUPlot in organizing and reading through large data sets, conducting a large number of calculations in seconds, and creating clear and informative graphics through tables and plots.

In all, this report provides valuable information and insight into the evolving pattern of the Great Lakes, along with thorough explanations of the implementation of C and GNUPlot used to present the data.

Question 1

The goal of this question was to calculate the yearly averages ice concentration percentages for each of the Great Lakes and the yearly concentration of all six lakes put together. Both the 2023 and 2024 averages were calculated and printed in two separate tables.

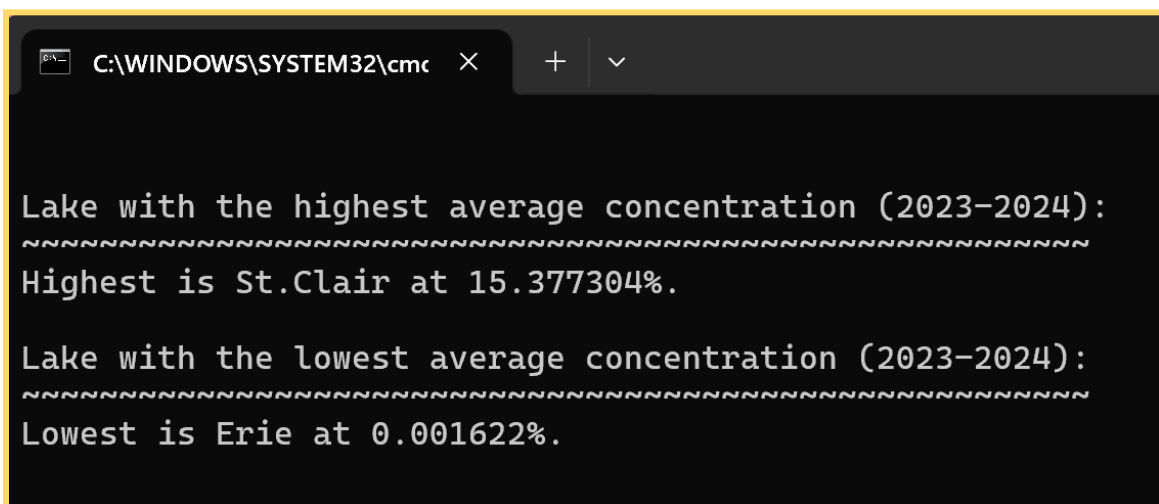
The 2023-2024 ice concentration data was put into a .txt file, which was opened and read by the program as an input file. A while loop was then used to read the columns of the file and store the values into a variable (year, day, ice concentration for each respective lake, and the total). Within the while loop, arrays were created in order to store all the data in each column associated with their respective variable. This was repeated until there were no more lines in the file to read, and the file was closed. Next, using for loops, the program read

Question 2

The goal of this question was to indicate what lake had the highest average ice concentration and what lake had the lowest average ice concentration within the 2023-2024 calculated average ice concentrations.

In order to complete this question, another array was made that stored all the average concentrations calculated in **Question 1**. Next, two variables for the highest and lowest values within the array were initialized to the zeroth row in the array. Variables for the line that contained said highest and lowest value were initialized to zero as well. Using a for loop, each line within the new array was read and compared with the highest and lowest variables. If the value of the i-th row in the array was bigger than the one stored in the highest variable, it was updated. Similarly, if the value of the i-th row in the array was smaller than the one stored in the lowest variable, it was updated. The line at which said highest and lowest value occurred was also updating simultaneously. This was done using if statements. These steps were repeated until there were no more rows to read within the array.

In order to print the lakes in which the highest and lowest average ice concentrations happened, a string was used that contained the names of all 6 lakes. They were doubled so that one set was associated with 2023 and the other set was associated with 2024. The purpose of this was the value for the highest and lowest line number would be used to call the string, which will in turn print the appropriate lake associated with the highest and lowest average concentration value.



```
C:\WINDOWS\SYSTEM32\cmd X + v

Lake with the highest average concentration (2023-2024):
~~~~~
Highest is St.Clair at 15.377304%.

Lake with the lowest average concentration (2023-2024):
~~~~~
Lowest is Erie at 0.001622%.
```

Question 2 Output

Question 3

The goal of this question was to indicate the day, year and highest ice concentration of each of the lakes. It was also important to take into account the possibility of a tie, which would mean both instances of the highest ice concentration would be printed.

A void function outside of the main program was used to complete this question. The void function utilized a similar method to **Question 2**, in which a variable for the highest ice concentration was initialized as the zeroth row in an array which contained the ice concentration data from the file. A for loop was used to ensure that each row in the array is read and compared. If the value of the i-th row in the array was bigger than the one stored in the variable, it was updated. This was done using if statements.

Within the void function, the highest level is printed along with the value that was stored in the highest ice concentration variable. In order to account for the possibility of a tie, another for loop along with if statements were used. If the value of the i-th row in the array was equal to the value stored in the variable, the year and day that it occurred would be printed utilizing the day and years array and putting the value of i as the row that would be printed. As the for loop runs, it will take into account all of the i-th rows whose value equals the value stored in the variable, which means all ties will be considered. The void function was called in the main program using the respective lake arrays, and the values were printed in a chart.

```

C:\WINDOWS\SYSTEM32\cmd  X  +  v

Highest Ice Concentration Percentage for Each Lake & When (2023-2024)
~~~~~
Lake Superior:
-----
Highest Level: 11.95%
Year: 2024, Day: 50

Lake Michigan:
-----
Highest Level: 18.28%
Year: 2024, Day: 21

Lake Huron:
-----
Highest Level: 22.65%
Year: 2024, Day: 22

Lake Erie:
-----
Highest Level: 35.72%
Year: 2024, Day: 22

Lake Ontario:
-----
Highest Level: 5.30%
Year: 2024, Day: 22

Lake St. Clair:
-----
Highest Level: 97.15%
Year: 2024, Day: 22
Year: 2024, Day: 23

```

Question 3 Output

Question 4

The goal of this question was to indicate which day, year and lake had the highest ice concentration in the entire data file. It was also important to take into account the possibility of a tie, which would mean both instances of the highest ice concentration would be printed.

Similarly to **Question 3**, a void function was used to complete this question. A 2D array was utilized in order to take into consideration both the rows and the columns of data within the array. The purpose of this was

Question 5

In order to create graphs that plot ice concentration per day, the data from the original file had to be sorted into a new indexed file. Indexing the data from the original file was crucial in creating a seamless graph because the days within the file reset to zero when the new year started. Consequently, the resulting plot had unwanted connecting lines.

The first step in creating the new indexed file for the plot was to open the file in writing mode. Then, a for loop was created with the parameters ($i=0; i \leq r; i++$). The index started at 0 because we want to iterate through all cells of our lake arrays. The second condition ensures that the loop prints all the values in the array; “r” is the integer that indicates how many rows of data we have. The last condition increments our index by 1. Hence, all cells of our lake arrays, 0 through “r”, are accounted for. The resulting file had 7 columns; the first column being the indexed days, and the rest of the columns being the data for each lake. After the loop, the file was closed.

GNUplot was used to plot our graphs. We uploaded the text file and plotted each graph accordingly. Each column of data in the file was graphed against the first column (our indexed days). We included axes labels, a title, a legend, and each line was coloured differently for each lake.

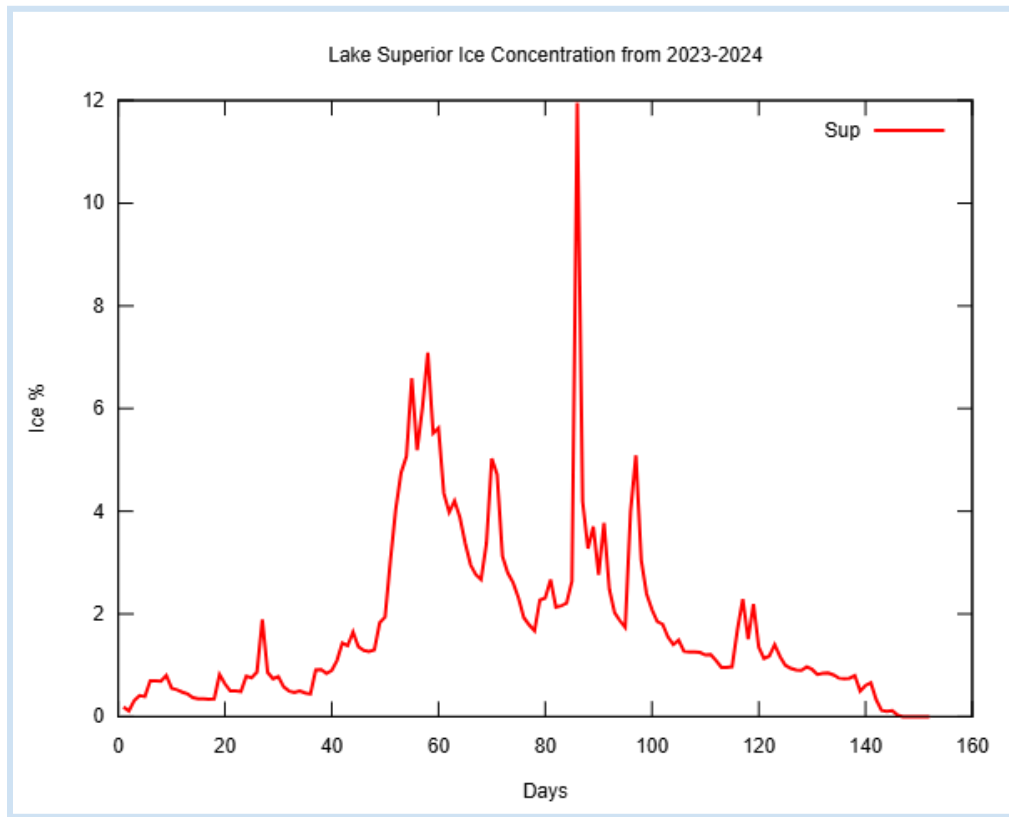


Figure 1: Plot of Lake Superior's Ice Concentration from 2023-2024.

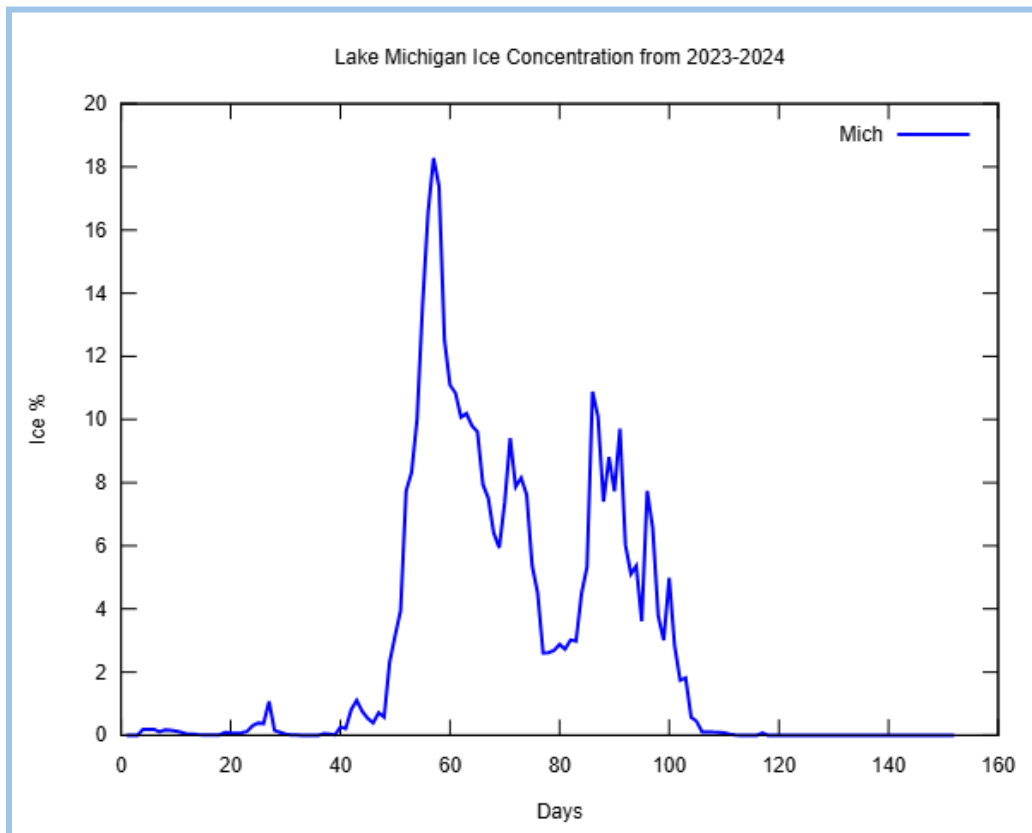


Figure 2: Plot of Lake Michigan's Ice Concentration from 2023-2024.

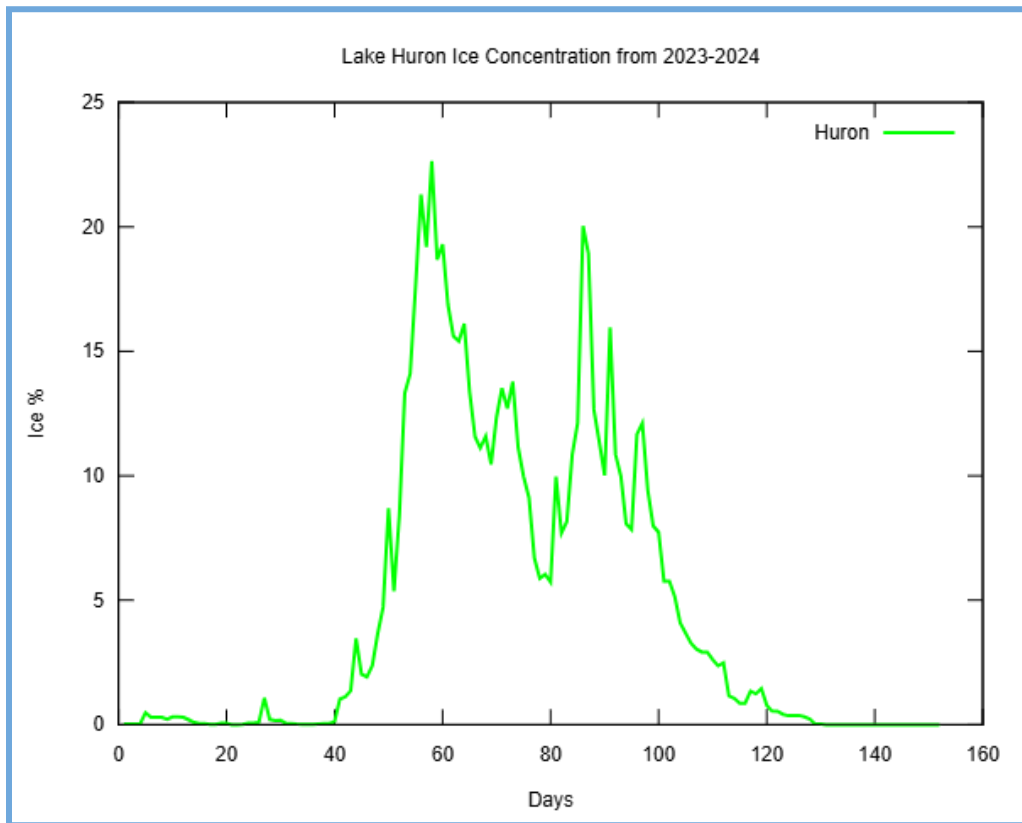


Figure 3: Plot of Lake Huron's Ice Concentration from 2023-2024.

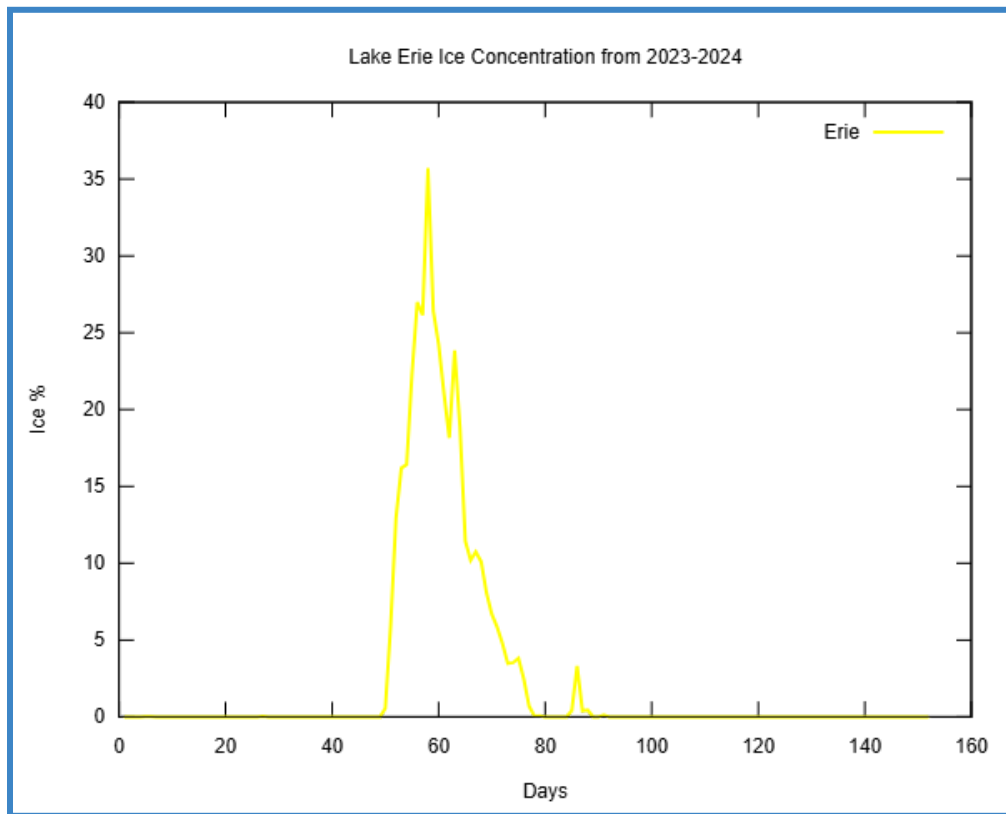


Figure 4: Plot of Lake Erie's Ice Concentration from 2023-2024.

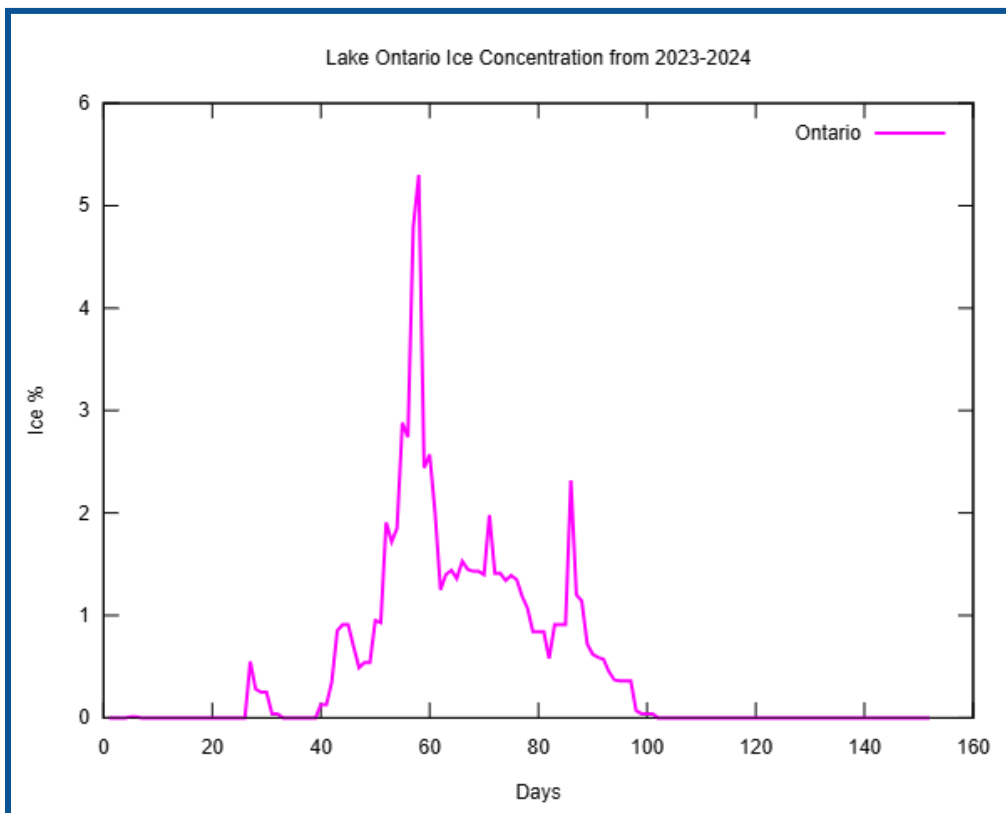


Figure 5: Plot of Lake Ontario's Ice Concentration from 2023-2024.

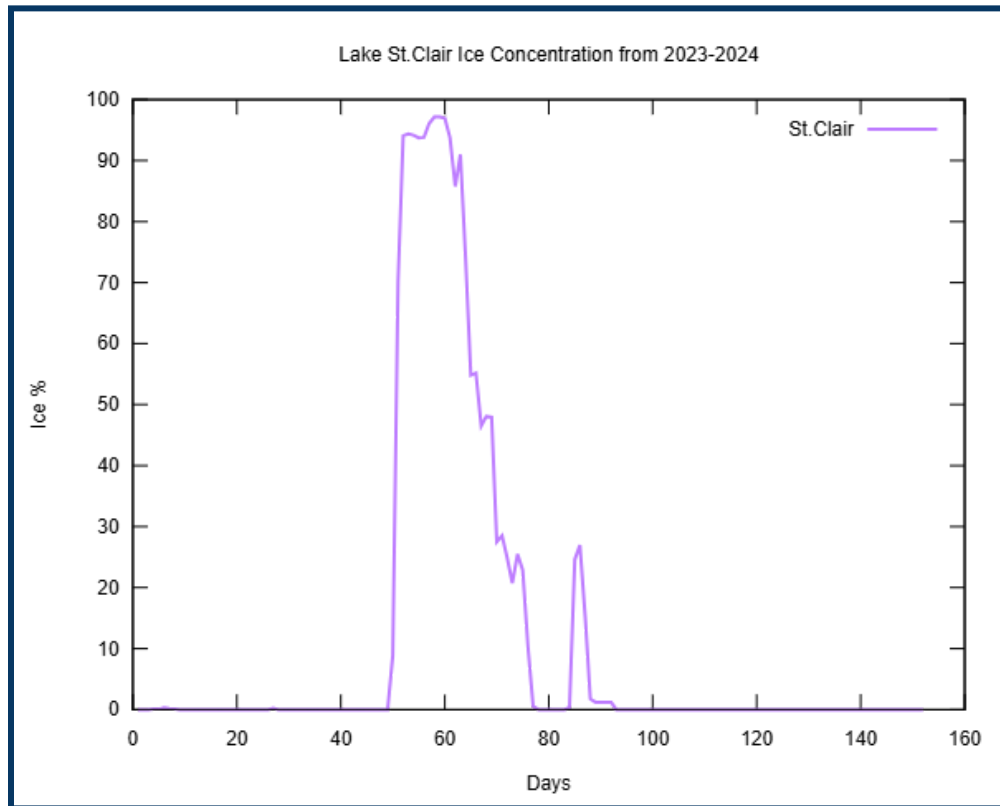


Figure 6: Plot of Lake St. Clair's Ice Concentration from 2023-2024.

Question 6

This question follows a similar process to the one in **Question 5**. Using the same file formed in the previous question, each column (lake) was plotted on one graph using GnuPlot. To ensure clarity, different colours were used for each lake and a legend was included in the top right of the plot.

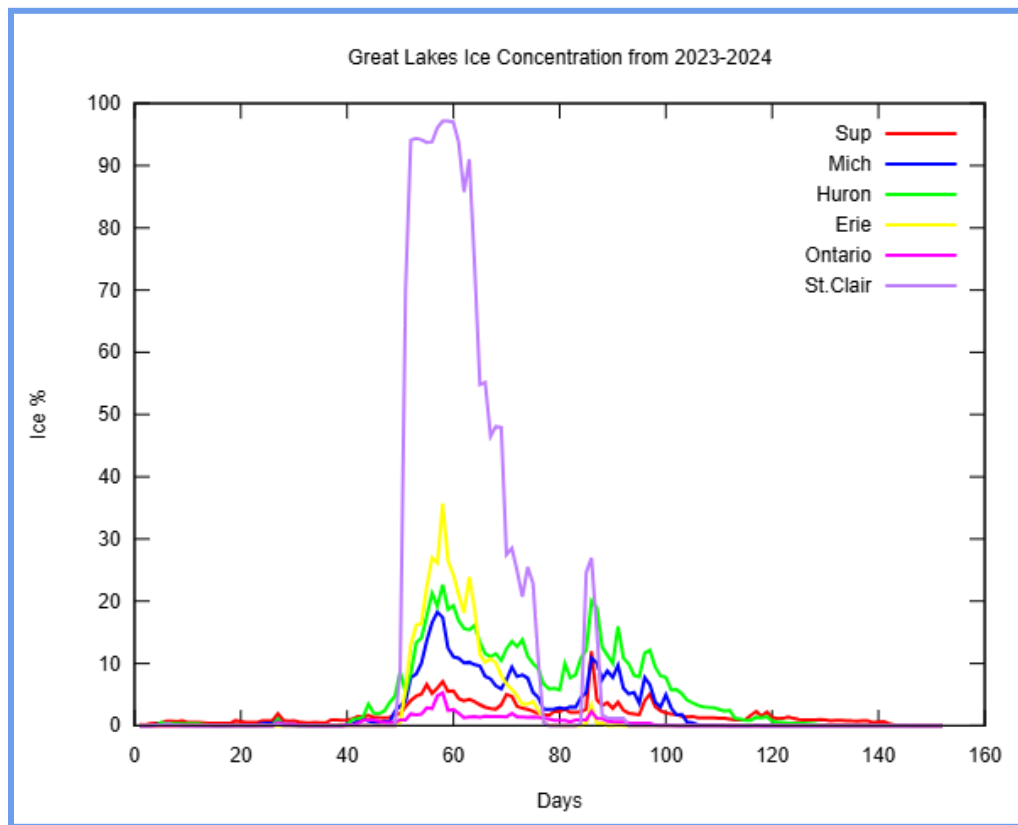


Figure 1: Plot of the Great Lakes Ice Concentration from 2023-2024.

Question 7

Since Question 7 repeats the process followed in **Question 1**, the same method was used to obtain the averages for the 2022-2023 and 2021-2022 data. There were numerous differences in the data between the years. Notably, St. Clair's ice concentration varied drastically between the years. In 2021, the ice concentration was 1.82%; a noticeable spike occurred, resulting in a concentration of 44% in 2022. The value dropped to 0.02% in 2023 and then raised to 15.4% in 2024. St. Clair's extreme variations may be due to it being a smaller lake compared to the others. Hence, St. Clair freezes over easier compared to larger lakes, like Michigan, when temperatures drop.

When comparing all the lakes between the years, the highest ice concentrations occurred during the first half of 2022. We can deduce that at that time there was a cold period that caused all the lakes to freeze more than normal. Although there were numerous variations in the data, as to be expected, the ones mentioned were the most notable.

```

C:\WINDOWS\SYSTEM32\cmd X + v

****2022-2023 GREAT LAKES ICE CONCENTRATION DATA****
~~~~~

2022 Great Lakes Ice Concentration Averages
~~~~~
Superior: 1.081163%
Michigan: 1.166512%
Huron: 1.438605%
Erie: 3.046047%
Ontario: 0.375349%
St.Clair: 16.840465%
Total: 1.379070%

2023 Great Lakes Ice Concentration Averages
~~~~~
Superior: 4.710426%
Michigan: 5.017021%
Huron: 8.655319%
Erie: 2.485674%
Ontario: 1.490000%
St.Clair: 9.329291%
Total: 5.293972%

```

Question 7A 2022-2023 Output

```

C:\WINDOWS\SYSTEM32\cmd X + v

Lake with the highest average concentration (2022-2023):
~~~~~
Highest is St.Clair at 16.840465%.

Lake with the lowest average concentration (2022-2023):
~~~~~
Lowest is Ontario at 0.375349%.

```

Question 7B 2022-2023 Output

```

C:\WINDOWS\SYSTEM32\cmd  ×  +  ∨

Highest Ice Concentration Percentage for Each Lake & When (2022-2023)
~~~~~
Lake Superior:
-----
Highest Level: 19.83%
Year: 2023, Day: 55

Lake Michigan:
-----
Highest Level: 21.51%
Year: 2023, Day: 35

Lake Huron:
-----
Highest Level: 34.37%
Year: 2023, Day: 55

Lake Erie:
-----
Highest Level: 40.18%
Year: 2023, Day: 36

Lake Ontario:
-----
Highest Level: 16.38%
Year: 2023, Day: 35

Lake St. Clair:
-----
Highest Level: 95.00%
Year: 2022, Day: 360
Year: 2022, Day: 361
Year: 2022, Day: 362

```

Question 7C 2022-2023 Output

```

C:\WINDOWS\SYSTEM32\cmd  X  +  v

Day, Year and Lake with the Highest Ice Concentration
~~~~~
Lake: St. Clair
Highest Level: 95.00%
Year: 2022, Day: 360
Year: 2022, Day: 361
Year: 2022, Day: 362

```

Question 7D 2022-2023 Output

```

C:\WINDOWS\SYSTEM32\cmd  X  +  v

****2021-2022 GREAT LAKES ICE CONCENTRATION DATA****
~~~~~

2021 Great Lakes Ice Concentration Averages
~~~~~
Superior: 1.247907%
Michigan: 1.352791%
Huron: 1.625116%
Erie: 0.082791%
Ontario: 0.300233%
St.Clair: 1.819535%
Total: 1.185581%

2022 Great Lakes Ice Concentration Averages
~~~~~
Superior: 22.545111%
Michigan: 12.589926%
Huron: 24.804148%
Erie: 26.232815%
Ontario: 5.664000%
St.Clair: 44.002519%
Total: 20.031704%

```

Question 7A 2021-2022 Output

```

C:\WINDOWS\SYSTEM32\cmd  X  +  v

Lake with the highest average concentration (2021-2022):
~~~~~
Highest is St.Clair at 44.002519%.

Lake with the lowest average concentration (2021-2022):
~~~~~
Lowest is Erie at 0.082791%.

```

Question 7B 2021-2022 Output

```

C:\WINDOWS\SYSTEM32\cmd  X  +  v

Highest Ice Concentration Percentage for Each Lake & When (2021-2022)
~~~~~
Lake Superior:
-----
Highest Level: 79.55%
Year: 2022, Day: 74

Lake Michigan:
-----
Highest Level: 37.47%
Year: 2022, Day: 45

Lake Huron:
-----
Highest Level: 67.23%
Year: 2022, Day: 57

Lake Erie:
-----
Highest Level: 93.76%
Year: 2022, Day: 31

Lake Ontario:
-----
Highest Level: 29.43%
Year: 2022, Day: 37

Lake St. Clair:
-----
Highest Level: 99.59%
Year: 2022, Day: 35

```

Question 7C 2021-2022 Output

```

Day, Year and Lake with the Highest Ice Concentration
~~~~~
Lake: St. Clair
Highest Level: 99.59%
Year: 2022, Day: 35

-----
(program exited with code: 0)

Press any key to continue . . .

```

Question 7D 2021-2022 Output**Question 8**

Once again, Question 8 used the same process followed in **Question 5** in order to index the data for daily ice concentrations and write it into another file. To get the monthly coverage trends for each year and each lake, a void function was written. The function takes in the input file, output file, and the “r” integer. To get the total number of months, we divide the total number of days by 30 (assuming each month has 30 days). To ensure the number of months is an integer and accounts for all the days, we use the ceiling function to round up to the nearest whole number. Both files are opened; one to write into and one to read from. Sum arrays for each lake and count array were initialized so each cell is 0.

Using a while loop, we obtain each daily average from each lake and add it to its respective sum array. It is important to note that at the beginning of the loop, we initialize the index to be an integer defined by “(day-1)/30”. In doing so, we can sum the average for each month under the index 0,1,2... and so on.

After the lake sum arrays are obtained, the averages for each month and each lake can be written into a new file using a for loop. The for loop condition is (int i = 0; i <n; i++). The index iterates until the last month as seen by the second parameter within the condition. Using an if statement within the for loop, ensures that we

are never dividing the sum by zero. Once that condition is satisfied, the month and each monthly average will be written into the file and then closed. This function was put into the main code for each dataset. Resulting in 3 files for the 3 different years.

Doing the same as previous questions, the files were uploaded to GNUplot and graphed accordingly. A legend and different colours for the lines were included to ensure clarity in interpreting the graph.

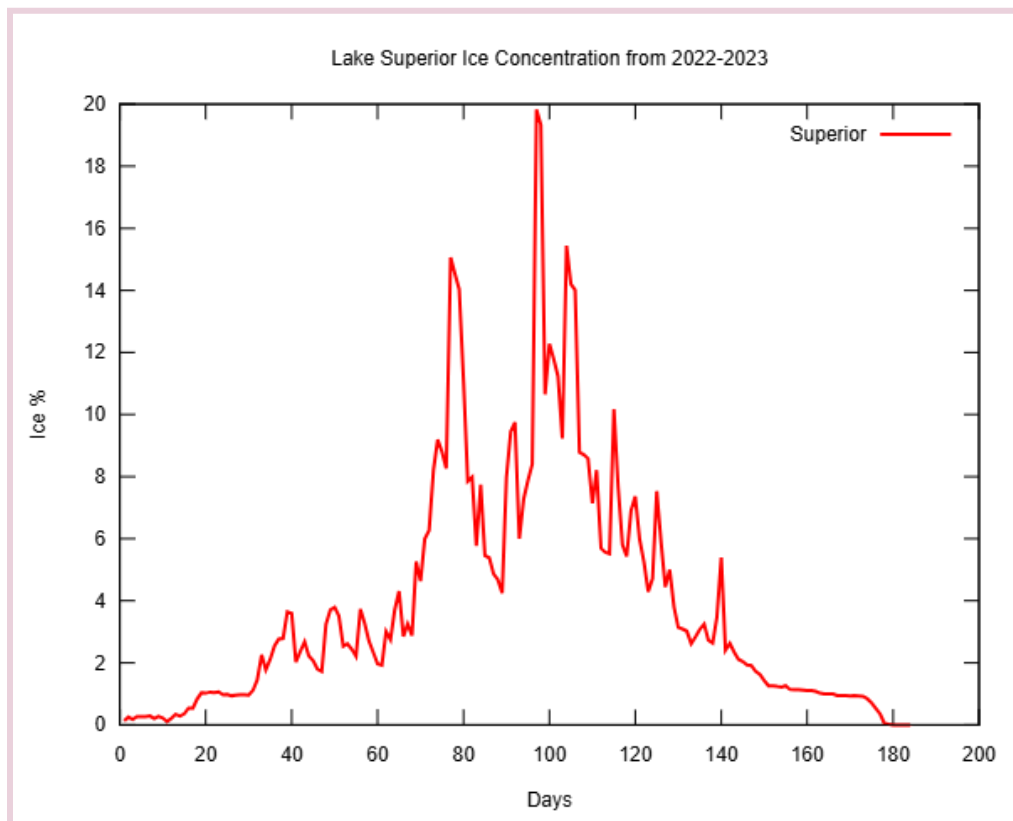


Figure 1A: Plot of Lake Superior's Ice Concentration from 2022-2023.

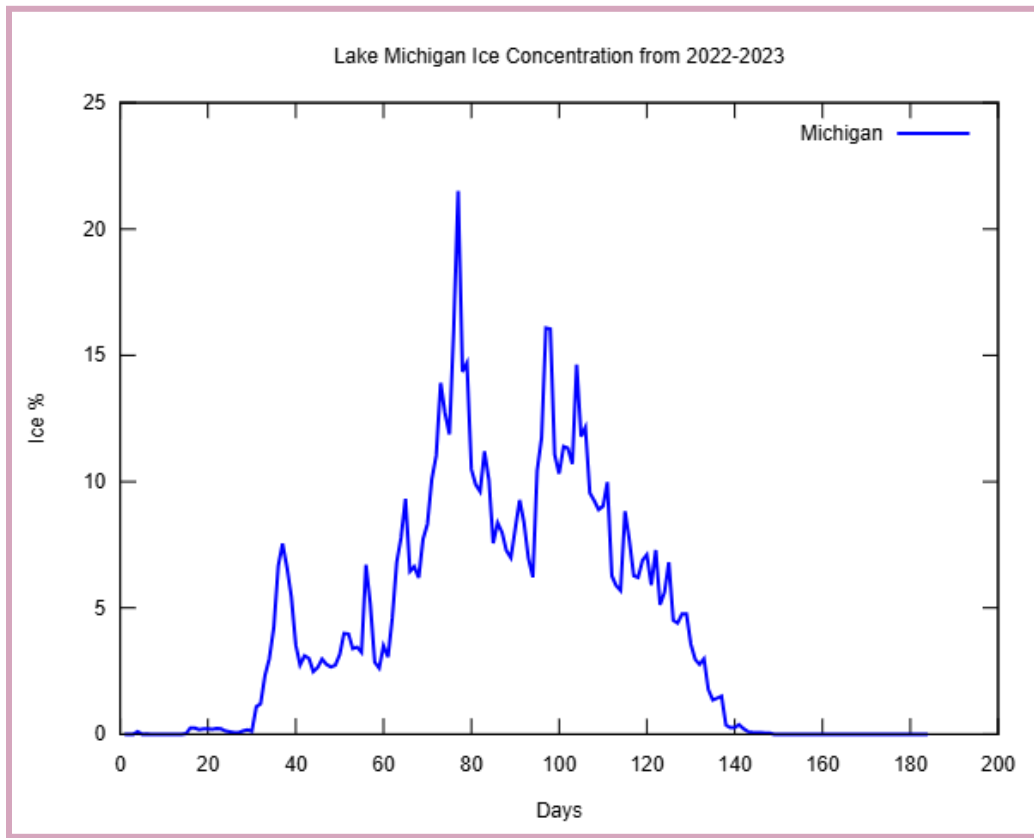


Figure 2A: Plot of Lake Michigan's Ice Concentration from 2022-2023.

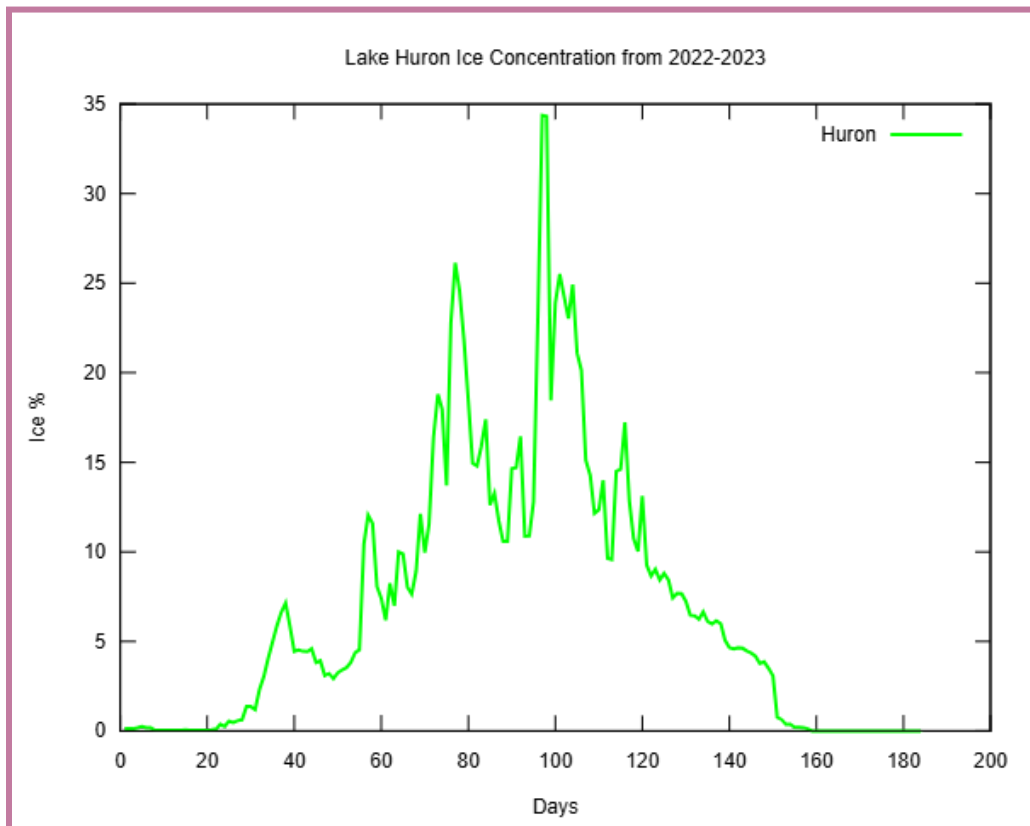


Figure 3A: Plot of Lake Huron's Ice Concentration from 2022-2023.

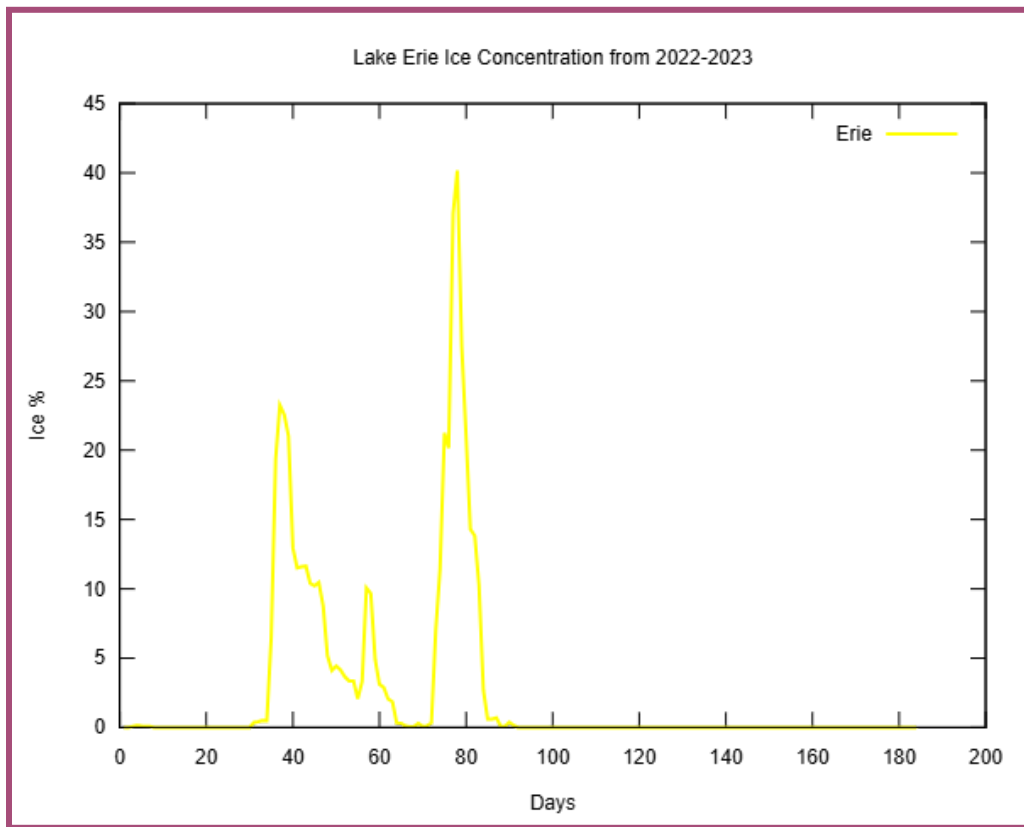


Figure 4A: Plot of Lake Erie's Ice Concentration from 2022-2023.

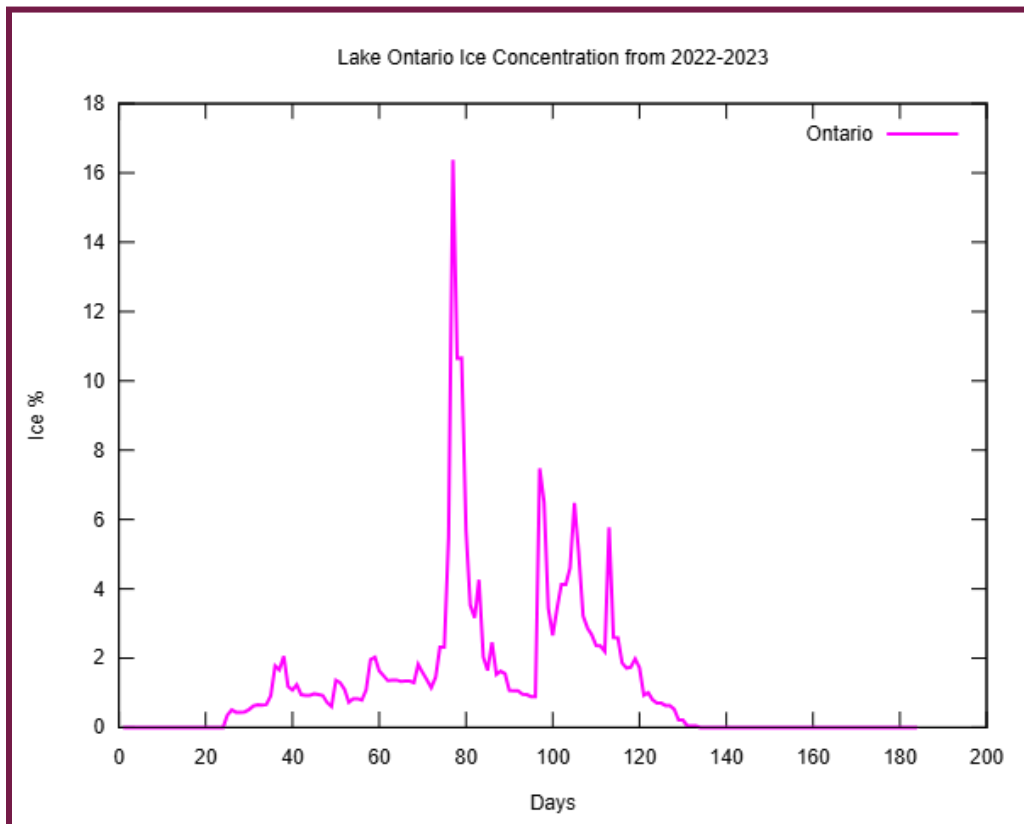


Figure 5A: Plot of Lake Ontario's Ice Concentration from 2022-2023.

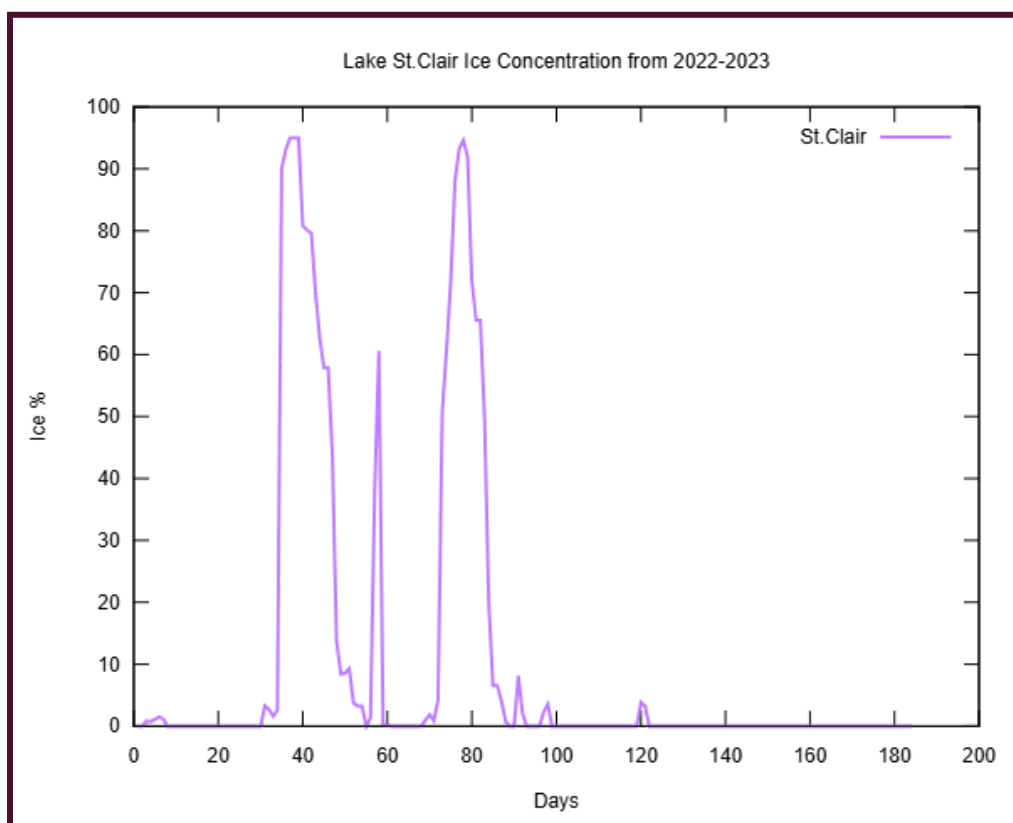


Figure 6A: Plot of Lake St. Clair's Ice Concentration from 2022-2023.

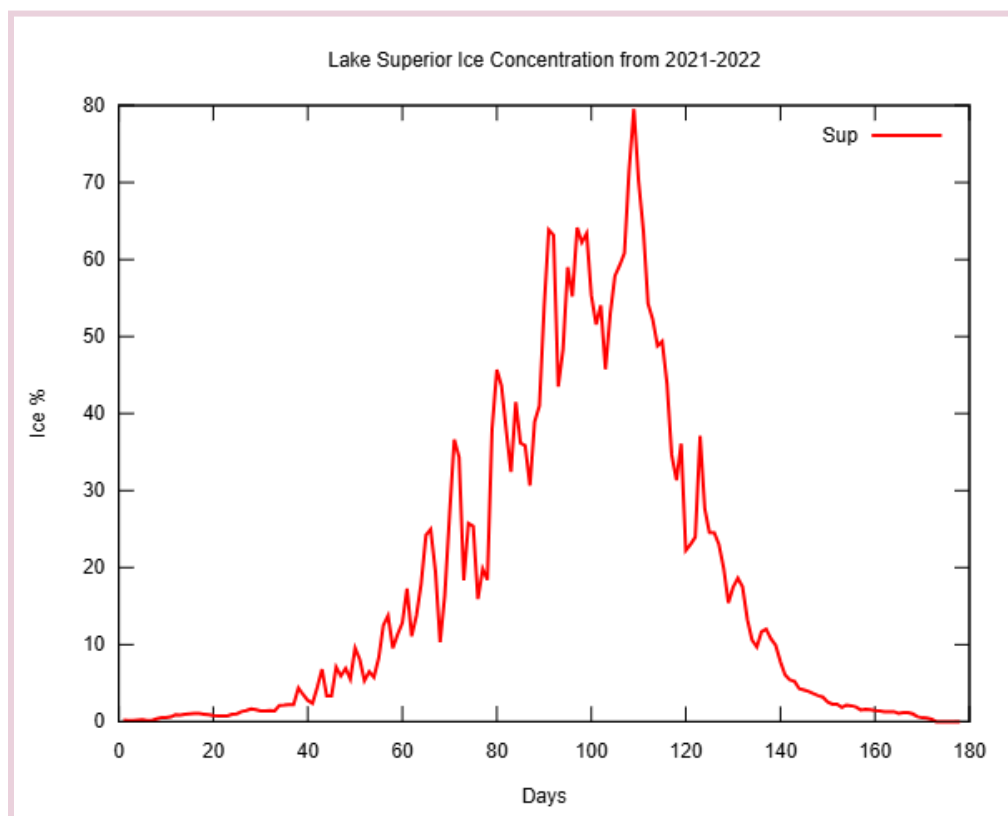


Figure 1B: Plot of Lake Superior's Ice Concentration from 2021-2022.

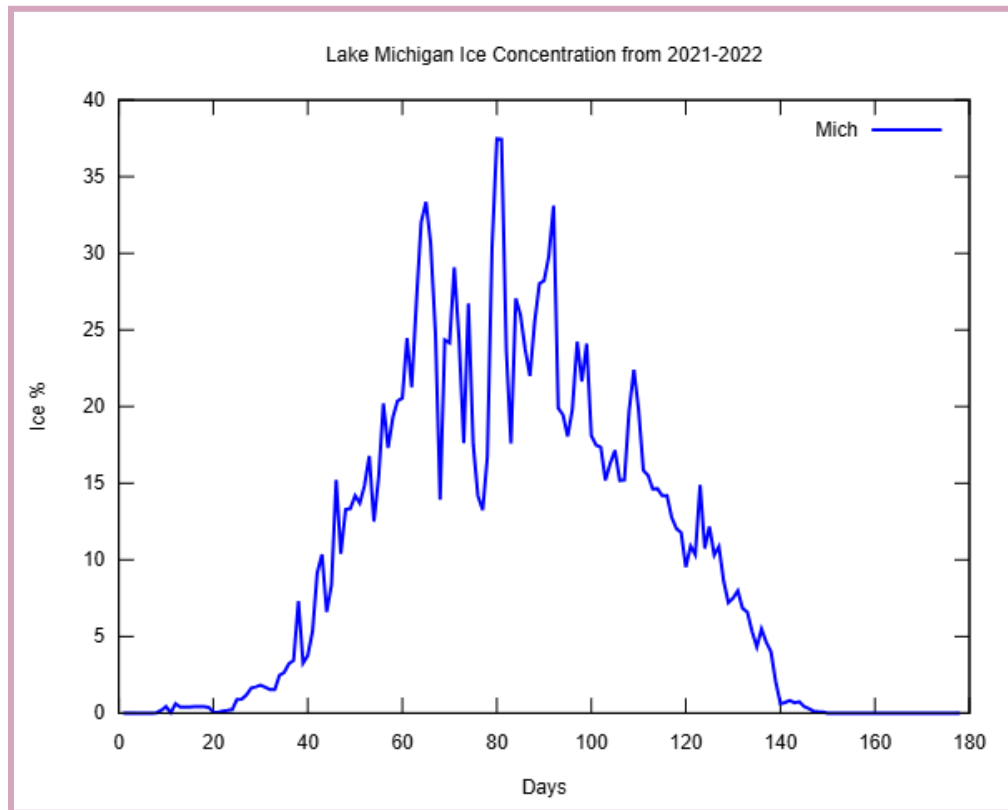


Figure 2B: Plot of Lake Michigan's Ice Concentration from 2021-2022.

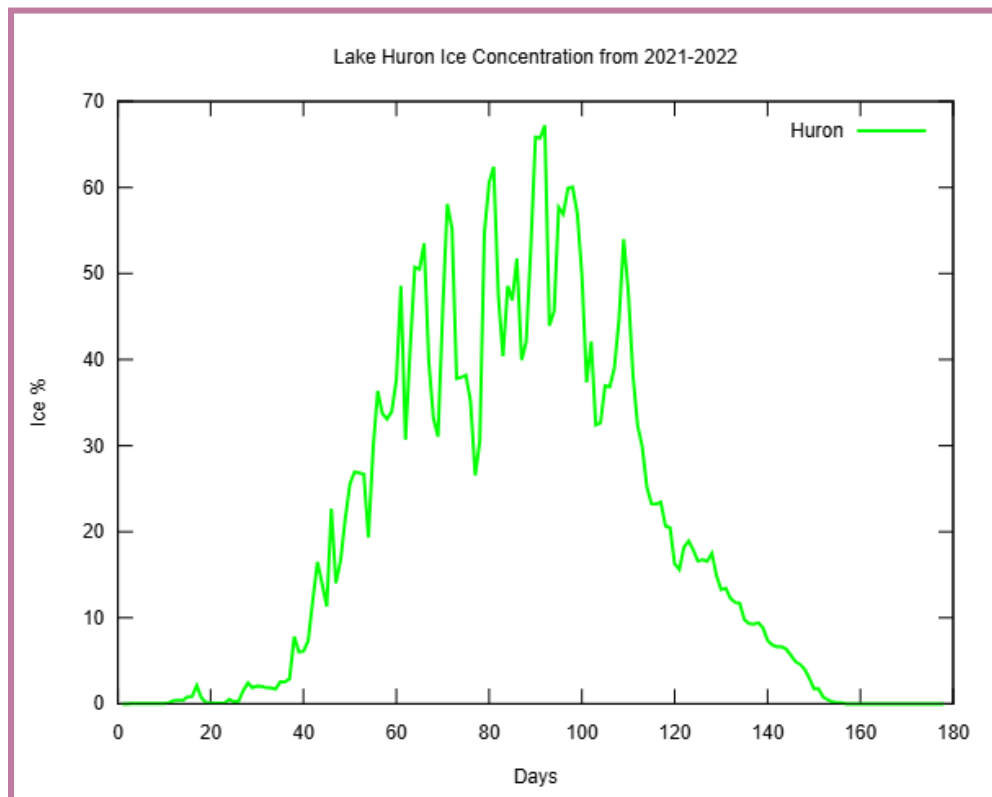


Figure 3B: Plot of Lake Huron's Ice Concentration from 2021-2022.

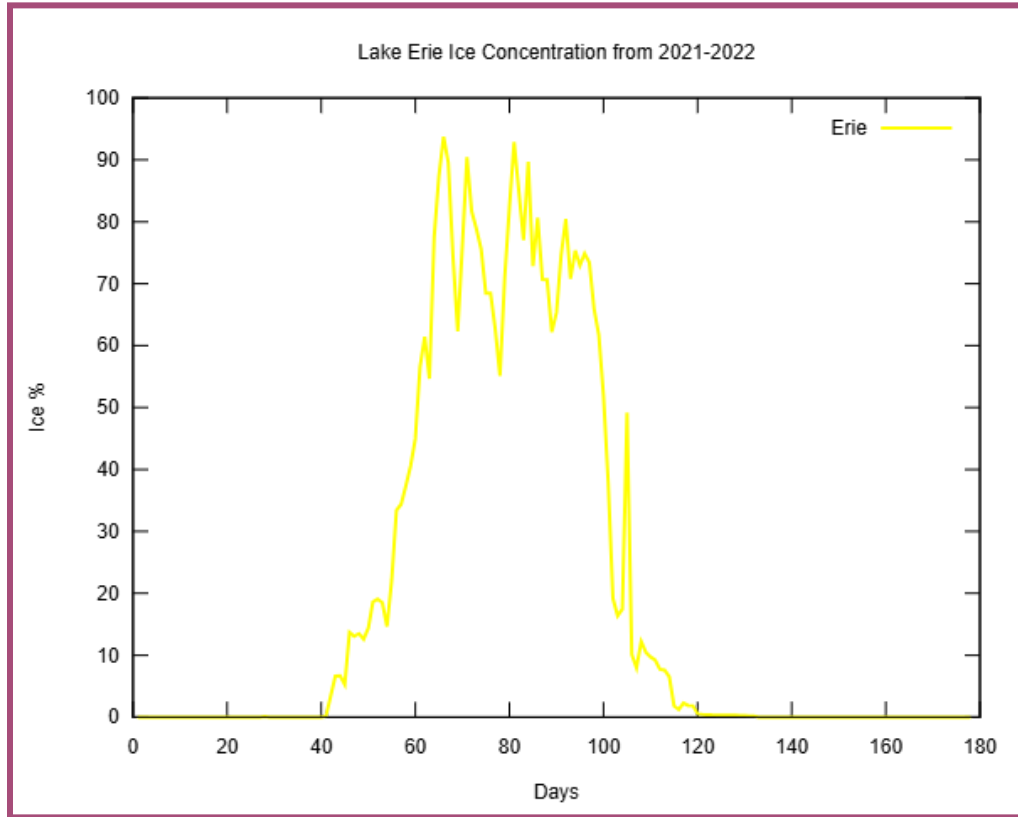


Figure 4B: Plot of Lake Erie's Ice Concentration from 2021-2022.

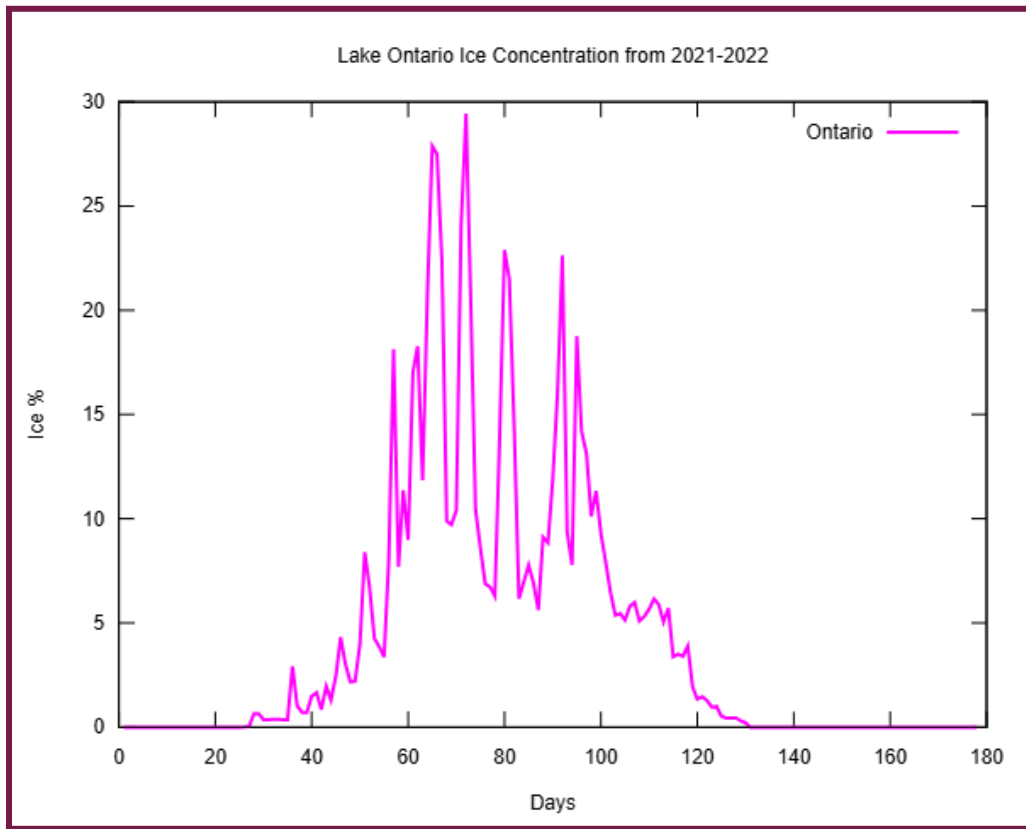


Figure 5B: Plot of Lake Ontario's Ice Concentration from 2021-2022.

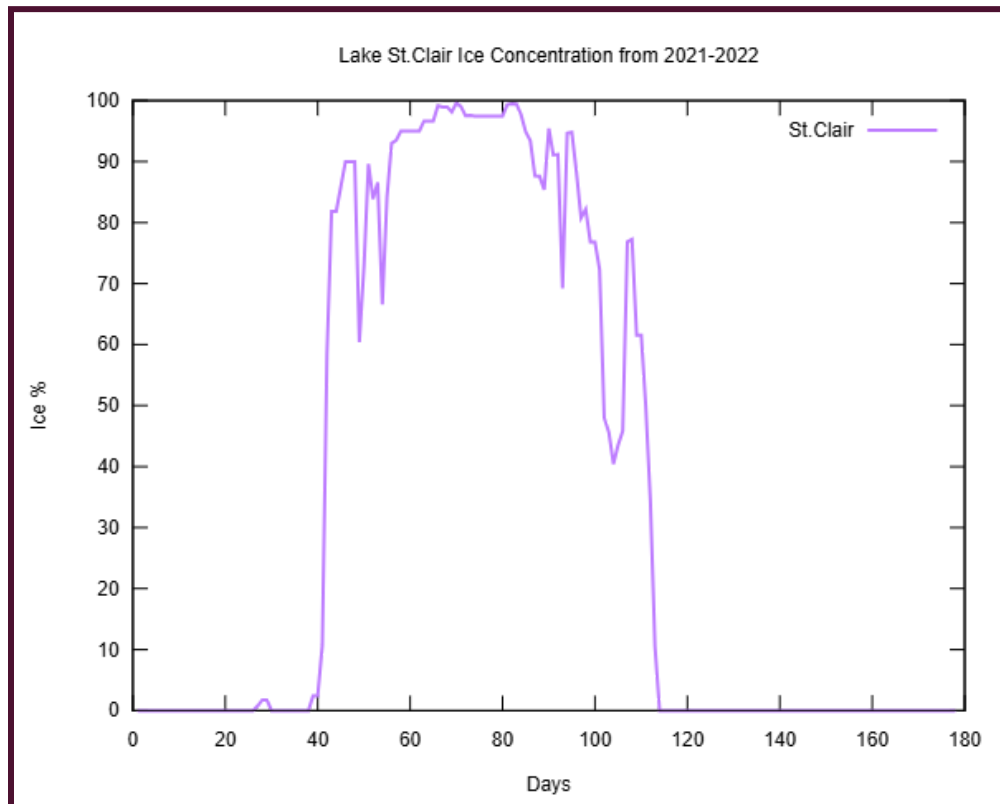


Figure 6B: Plot of Lake St. Clair's Ice Concentration from 2021-2022.

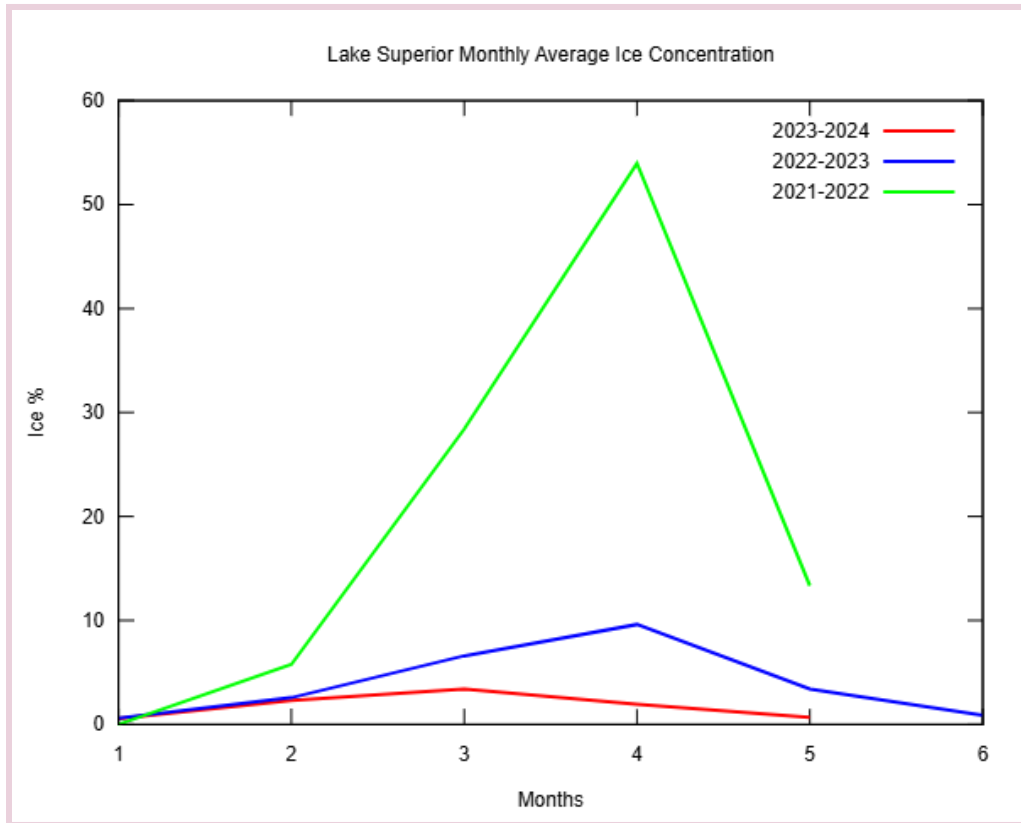


Figure 1C: Plot of Lake Superior Monthly Average Ice Concentration for Each Year.

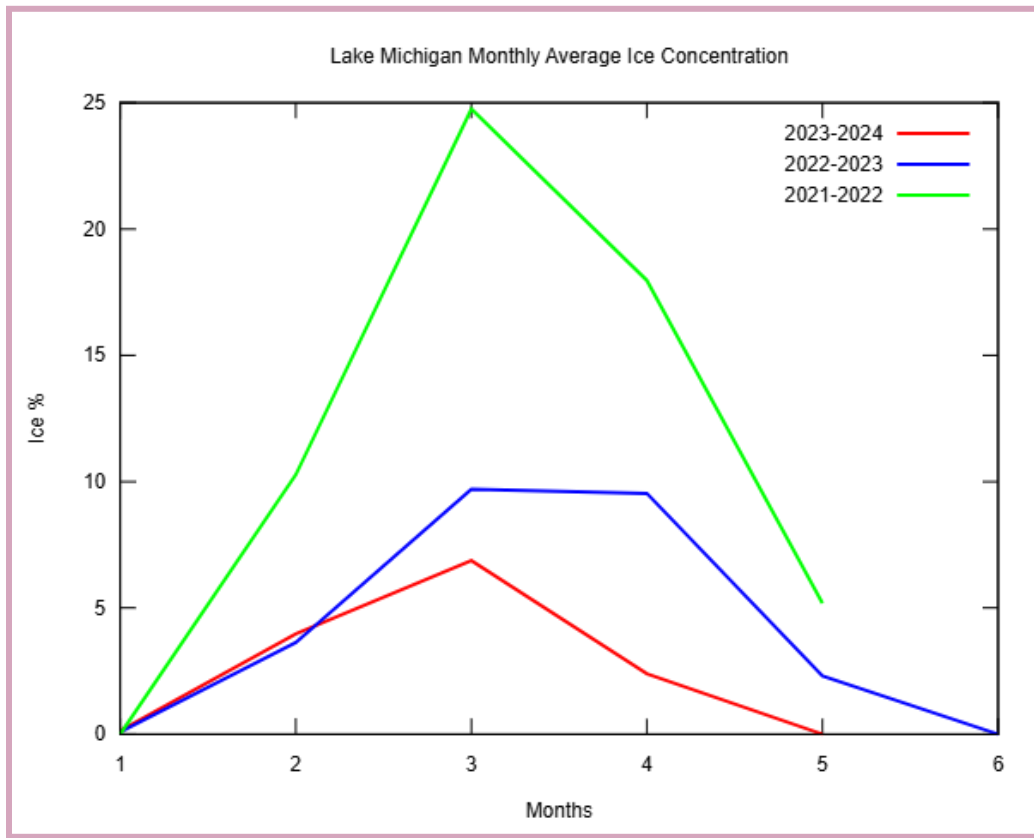


Figure 2C: Plot of Lake Michigan Monthly Average Ice Concentration for Each Year.

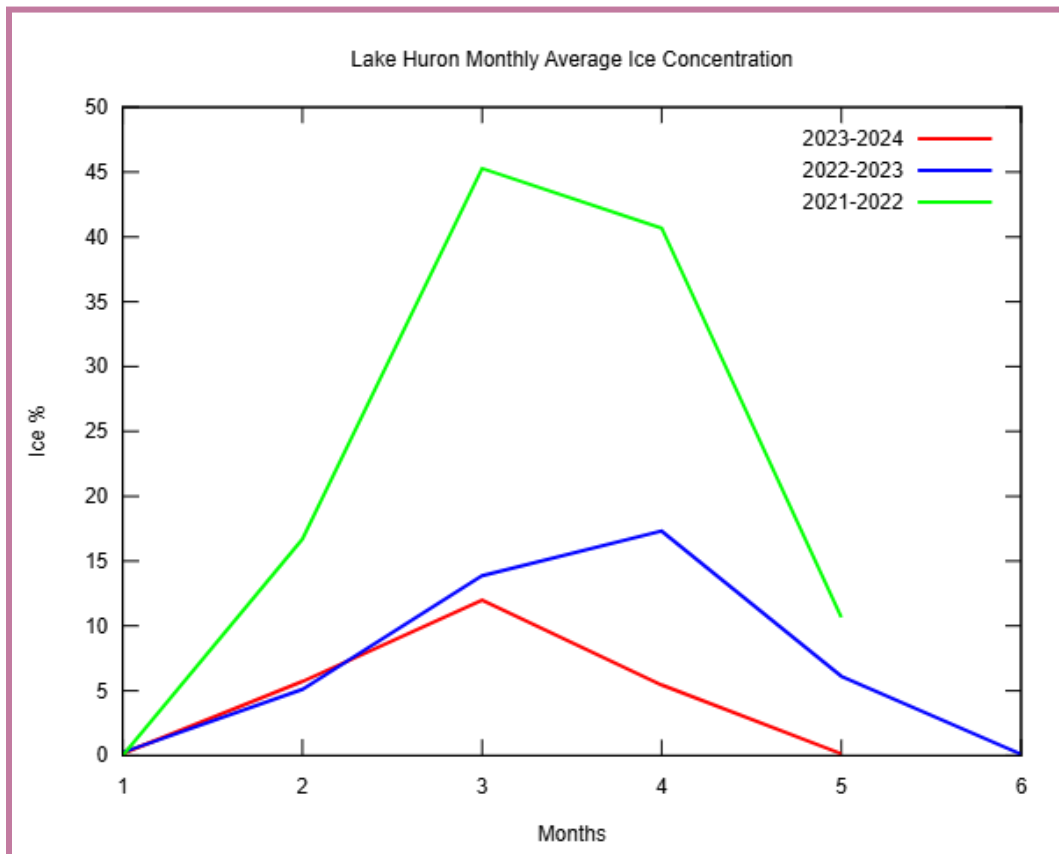


Figure 3C: Plot of Lake Huron Monthly Average Ice Concentration for Each Year.

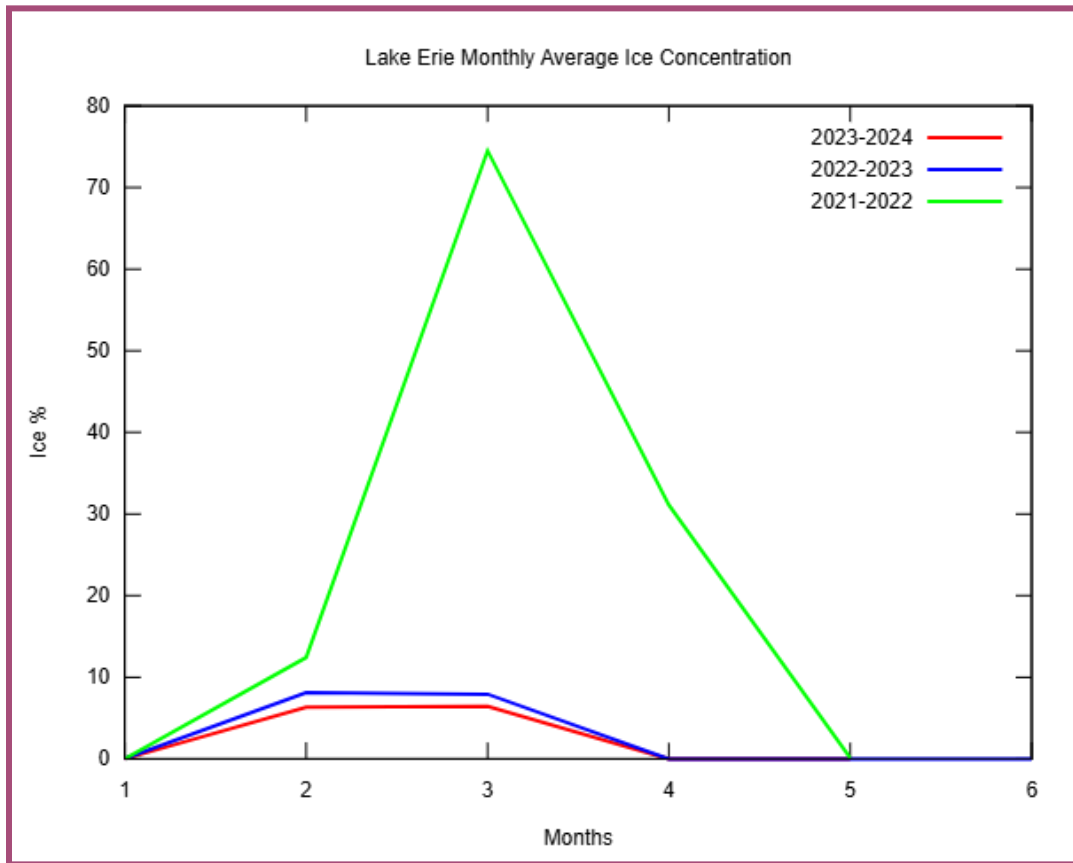


Figure 4C: Plot of Lake Erie Monthly Average Ice Concentration for Each Year.

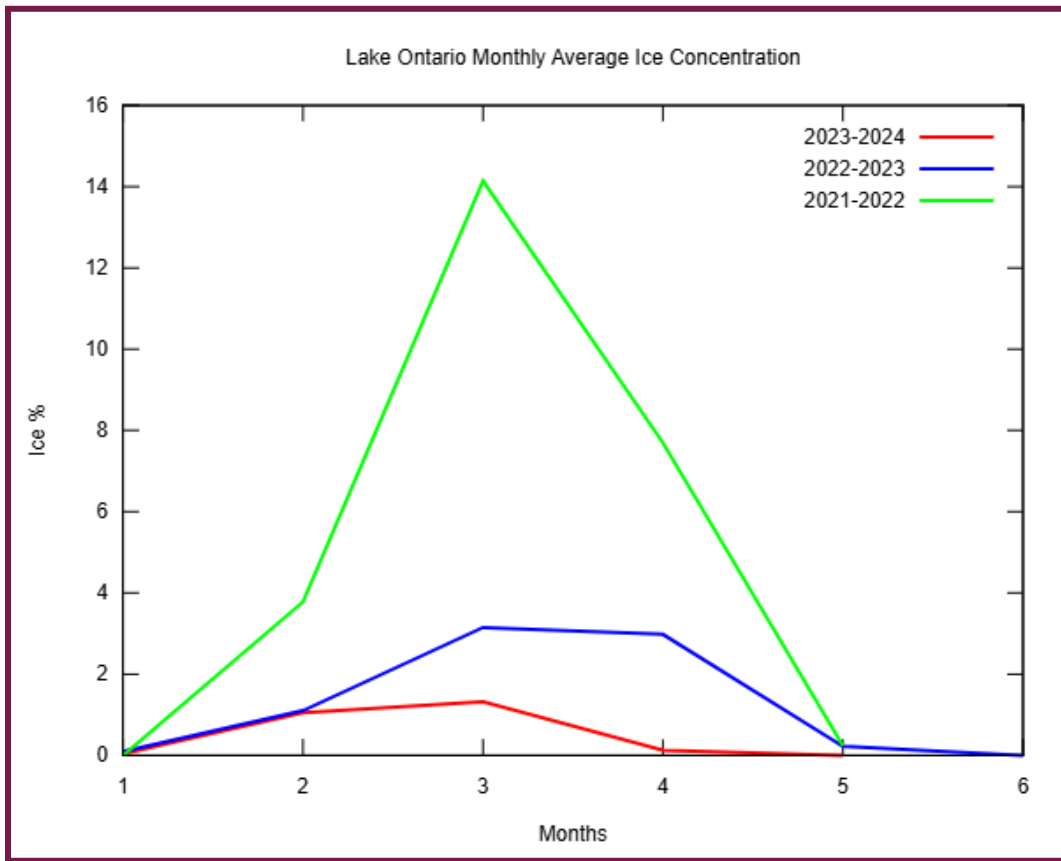


Figure 5C: Plot of Lake Ontario Monthly Average Ice Concentration for Each Year.

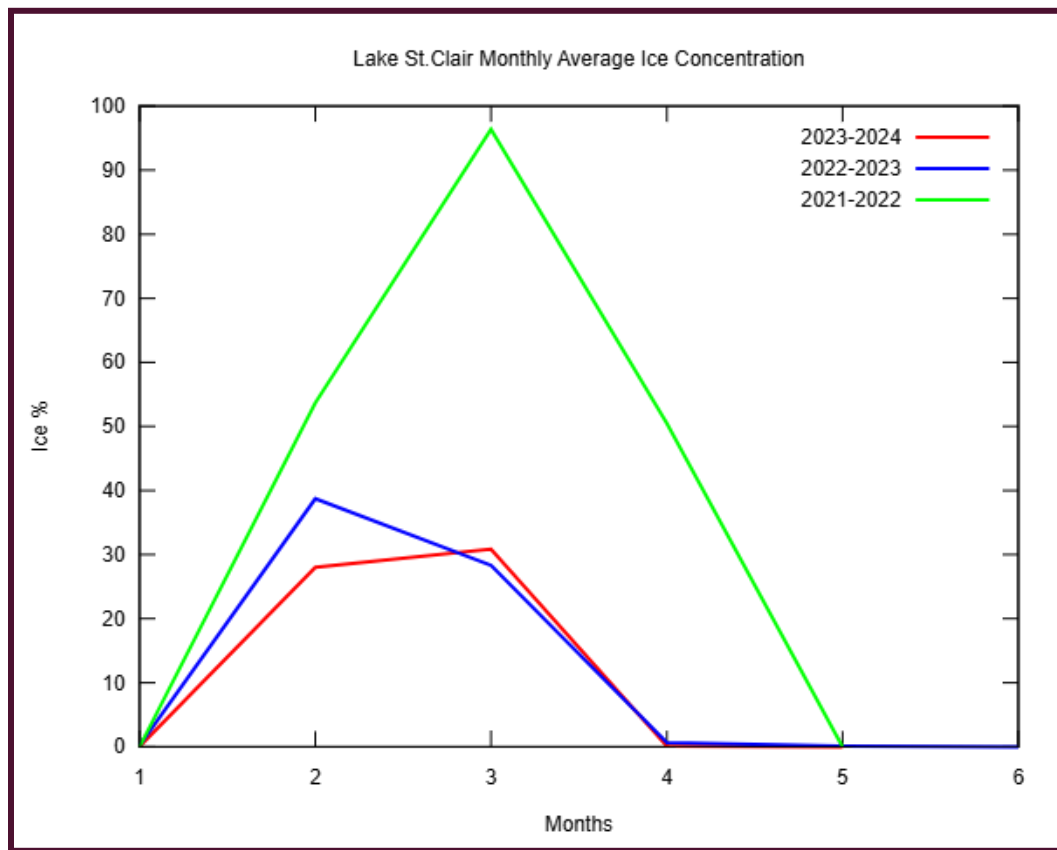


Figure 6C: Plot of Lake St. Clair Monthly Average Ice Concentration for Each Year.

Conclusion

Through organizing and graphing the Great Lake ice concentration data using C language and GNUplot, we strengthened our coding and problem solving skills. Working with such a large set of data aided us in creating more “vague” functions that can be reused for multiple different sets of data.

Although we learned a lot from this project, if we were to do it again, we would try to be more concise with our variables from the beginning of the project. When we first started writing the code, we underestimated how many similar variables would be needed. For example, we needed arrays for each lake each year. We did not account for this from the beginning which caused us to get confused with the numerous variables we had. Organization in general was a challenge for us due to how extensive the project was. For our next projects, we would include more comments for each step to minimize confusion among ourselves and other people who may review our code.

Overall, this project served as a crucial step in strengthening our C programming skills and critical thinking skills when it comes to solving complex problems.